

Project Proposal

ClinicBuddy AI: A Multi-Agent Patient Communication and Clinical Visit Preparation Assistant

1. Problem Statement & Motivation

- Effective communication between patients and healthcare providers is the most essential key for accurate clinical assessment. However, many patients still find it difficult to clearly describe their symptoms during medical visits due to nervousness, limited medical vocabulary, or language barriers. Especially the elderly and patients with limited English proficiency are affected, often seeing incomplete or unclear system descriptions that reduce consultation efficiency and increase the likelihood of misunderstanding.
- Existing patient portals on the market right now are just focusing on passive systems, such as storing health records, displaying laboratory results, and enabling appointment scheduling or messaging to the caregivers. They do not actively help patients to explain the structure of their health concerns before the clinical appointment. From that, we can see there is still an important gap in healthcare communication support technologies.
- ClinicBuddy AI will help to solve this problem by providing an intelligent visit-preparation assistant that can guide patients through structured symptom interviews, clarifying incomplete information, support multilingual communication, and generate a concise clinician summary for the clinician to improve patient-provider communication quality.

2. Target Users

- ClinicBuddy AI is focused on two primary user groups:
 - a. **Patients:** Individuals who are unsure how to explain symptoms. Elderly patients require guided communication support. Patients with limited English proficiency need translation assistance.
 - b. **Clinicians:** Healthcare providers who must review patients' histories and require concise summaries, and they need a summary and risk indicators to support decision-making.
- ⇒ *This system does not provide medical diagnoses or treatment recommendations.*

3. Why is an Agentic AI Solution Appropriate?

- What makes ClinicBuddy AI apart from other traditional patient portals is its introduction of an intelligent communication preparation agent to help patients organize and articulate their health concerns before their appointment. The system transforms patients' unstructured thoughts into a concise summary, ready for doctors to review.

Simultaneously, it assists in clarifying doctors' questions and patients' answers through reasoning and multilingual translation.

- The multi-agent system allows different specialized agents to control each stage independently and also ensures they all collaborate in a coordinated workflow. By focusing on communication intelligence rather than diagnosis, this system represents a novel application of agent-based Artificial Intelligence to create greater clarity, completeness, and accessibility in patient-doctor interactions.

4. Use Cases and Scenarios

a. Scenario 1: Symptom Interview and Structuring

- A patient interacts with ClinicBuddy AI before a medical visit. The system asks about the patient's symptoms by guided questions and, based on that, identifies missing details, and produces a structured "Doctor Brief" which summarizes key symptom information.

b. Scenario 2: Clinical Review

- A clinician selects a patient profile and views a dashboard showing trends over time. The system is generating a full, informative clinical summary highlighting recent changes and risk flags.

c. Scenario 3: Safety Verification

- If a patient asks health-related questions, such as "Is my heart rare in the dangerous condition?", the system just provides educational information while avoiding medical diagnosis and treatment advice to ensure the safety of the agent.

d. Scenario 4: Multilingual Communication Support

- Patients can describe symptoms in their preferred language. The system translates the information and produces both patient-language scripts and English clinician summaries.

5. Initial Systems Design

- ClinicBuddy AI consists of five specialized agents:

Agent	Role
1/ Data Intake Agent / Interview Agent	Confirms and stores patients' health data by conducting guided patient symptom interviews.
2/ Structuring & Trend Analysis Agent	Analyze the patient's historical signs and generate trends graph to show if the condition is improving, stable, or worsening based on recent data. After that, the agent will generate ready summaries for clinicians to use.
3/ Translation Agent	Support multilingual communication and language conversion.
4/ Clarification Agent	Detect missing or vague information and request follow-up details.

5/ Safety Agent	Reviews outputs to prevent diagnosis or unsafe medical advice.
-----------------	--

- Tool and Data Sources:

- Python functions for data validation and analysis.
- Local database for storing and controlling patient records.
- LLM API for natural language generation.
- Streamlit for the web-based front-end interface and built-in charts.

6. Feasibility and Risks

a. Potential Challenges:

- Language models might generate inaccurate or misleading information.
- Output may lead to medical advice.
- Simulated data cannot reflect real-world complexity.
- Translation or summarization inaccuracies.

b. Mitigation Strategies:

- Restrict output to just explanations and summaries only.
 - Implement a Safety agent and test all the functions to filter unsafe outputs.
 - Use rule-based risk detection while developing predictive models.
- ⇒ *The effectiveness of the system will be evaluated by comparing patient-written symptom descriptions with structured medical summaries generated by artificial intelligence, while also measuring improvements in completeness, clarity, and user trustworthiness.*

7. Project Timeline

- Proposal: Define problem and system design
- Milestone I: Implement front-end interface
- Milestone II: Integrate multi-agent workflow
- Milestone III: Conduct evaluation and pilot testing
- Final Phase: Record demo video and submit final report